F12 4th CAR

Lecture 15: Section 2.4
Complex Numbers

Imaginary unit

Complex numbers

Operations with complex numbers

Complex conjugate

Rationalize the denominator

Square roots of negative numbers

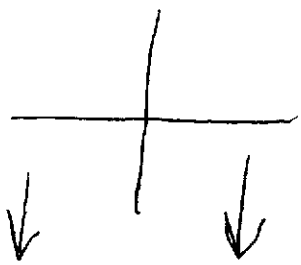
Complex solutions of quadratic equations

Graph. $f(x) = -x(x-3)^2(x+2)$ **Practice**

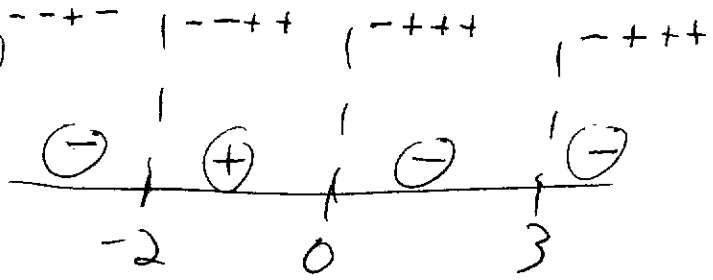
① Leading coef: $-x^4$

② zeros: 0, 3, −2

multiplicity: odd, even, odd

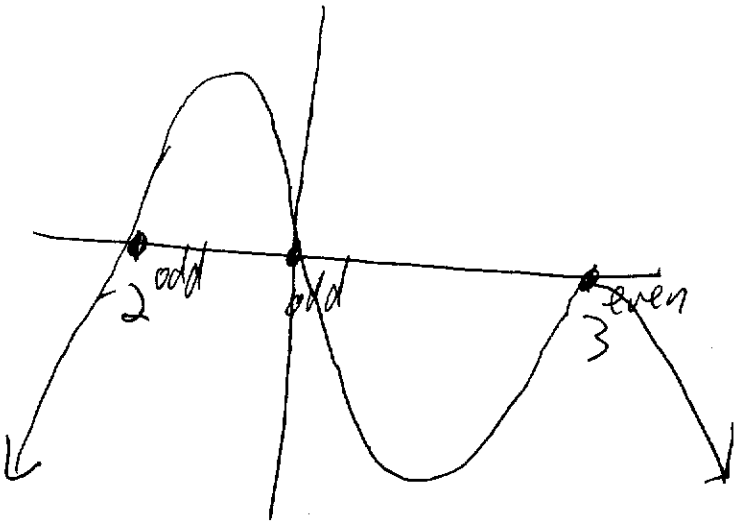


③



④ y-int

$f(0) = 0$



4th

$$\frac{3x^4-6x^2-12x-4}{x-3} = 3x^3+9x^2+21x+51+\frac{149}{x-3}$$

The Remainder Theorem Find remainder when dividing by $x-3$,

If a polynomial is divided by $x-k$, then the remainder is $r = f(k)$.

$f(x) = 3x^4-6x^2-12x-4$

$f(3) = \cdots = 149$

The Factor Theorem

A polynomial $f(x)$ has a factor $x-k$ if and only if $f(k) = 0$.

ex. Given $f(x) = x^5 - x + 1$.

1) Evaluate $f(1)$. $= 1^5 - 1 + 1 = 1 - 1 + 1 = 1$

~~2) Use the Remainder Theorem to evaluate $f(1)$.~~

3) Is $x-1$ a factor of f? ~~No since $f(1)$~~

No, since $f(1) \neq 0$.

Sakai

ex. Show that $x - 1$ and $x + 1$ are factors of $f(x) = x^4 + 5x^3 + 5x^2 - 5x - 6$. Then factor $f(x)$ completely.

Consider the equation $x^2 = -1$. No real solution, however

Def. The **imaginary unit**, i, is the number such that

$$* \quad \boxed{i^2 = -1 \quad \text{or} \quad i = \sqrt{-1}} \qquad \boxed{i^4 = 1} \quad *$$

$= \boxed{-i}$ $\qquad = -1 \cdot -1 = \boxed{1}$

$= -1 \cdot i$

Power of i

$$* \quad \boxed{i^1 = \sqrt{-1} = i} \quad * \quad \boxed{i^2 = -1} \qquad i^3 = i^2 \cdot i = \qquad i^4 = i^2 \cdot i^2 =$$

$$i^5 = i^4 \cdot i = 1 \cdot i \qquad i^6 = i^4 \cdot i^2 \qquad i^7 = i^4 \cdot i^3 \qquad i^8 = i^4 \cdot i^4 = 1 \cdot 1 = 1$$

$= \boxed{i} \qquad = 1(-1) = -1 \qquad = 1 \cdot (-i) = \boxed{-i}$

Therefore, every integer power of i can be written as $i,\ -1,\ -i,\ 1$.

In general, divide the exponent by 4 and rewrite:

ex. 1) i^{85} $\qquad$ 2) $(-i)^{85} = -(i^{85}) = \boxed{-i}$

$= (i^4)^{21} \cdot i = 1^{21} \cdot i = \boxed{i}$

3) i^{100} $\qquad$ 4) $(-i)^{-18} = \dfrac{1}{(-i)^{18}} = \dfrac{1}{i^{18}}$

$(i^4)^{25} = 1^{25} = \boxed{1}$

$= \dfrac{1}{(i^4)^4 \cdot i^2} = \dfrac{1}{1^4 \cdot (-1)} =$

$= -1$

Def. **Complex numbers** are numbers of the form $a + bi$, where a and b are real numbers.

a is the **real part** and b is the **imaginary part** of the complex number $a + bi$.

$a + bi$ is called the **standard form** of a complex number.

ex. Write the number -5 as a complex number.

$-5 = \boxed{-5 + 0i}$

Complex #s

Real #s Imaginary

NOTE: The set of real numbers is a subset of the set of complex numbers.

If $b = 0$, the number $a + 0i = a$ is a real number.

If $a = 0$, the number $0 + bi = bi$, is called a **pure imaginary number**.

Equality of Complex Numbers

$a + bi = c + di$ if and only if $a = c$ and $b = d$.

Operations with Complex Numbers

Sum: $(a+bi)+(c+di) = \boxed{(a+c)+(b+d)i}$

Difference: $(a+bi)-(c+di) = a+bi-c-di =$

Multiplication: $(a+bi)(c+di) = \boxed{(a-c)+(b-d)i}$

FOIL: $= ac + adi + bci + bdi^2$ $\rightarrow -1$

$= ac + adi + bci - bd$

$= \boxed{(ac-bd)+(ad+bc)i}$

NOTE: Use the distributive property (FOIL) and remember that $i^2 = -1$.

<u>ex.</u> Write in standard form:

1) $3(2-5i)-(4-6i) =$

$6 - 15i - 4 + 6i =$

$(6-4)+(-15+6)i = \boxed{2-9i}$

2) $(2+3i)(4+5i)$

FOIL $= 8 + 10i + 12i + 15i^2$ $= -1$

$= 8 + 10i + 12i - 15$

$= (8-15) + (10+12)i = \boxed{-7+22i}$

Note: Many books (however not ours) use a bar to signify conjugate.

Complex Conjugates

Def. The **conjugate (complex conjugate)** of a complex number $z = a + bi$ is $\overline{z} = \overline{a + bi} = \boxed{a - bi}$

ex. Find the conjugate of the following:

1) $-1 - 3i$ conjugate is $\boxed{-1 + 3i}$

2) $-2i = 0 - 2i$, conj $\rightarrow 0 + 2i = \boxed{2i}$

3) $5 = 5 + 0i$, conj $\rightarrow 5 - 0i = \boxed{5}$

NOTE: A real number is its own conjugate.

Operations with Conjugates

1. $z + \overline{z} = (a+bi) + (a-bi) = a + \cancel{bi} + a - \cancel{bi} = \boxed{2a}$
2. $z - \overline{z} = (a+bi) - (a-bi) = \cancel{a} + bi - \cancel{a} + bi = \boxed{2bi}$
3. $z\overline{z} = (a+bi)(a-bi) = a^2 - \cancel{abi} + \cancel{abi} - b^2 i^2$ $i^2 = -1$

FOIL * memorize

$\boxed{= a^2 + b^2}$

Theorem: The product of a complex number $z = a + bi$ and its conjugate is a nonnegative real number:

$$z\overline{z} = a^2 + b^2$$

Rationalize the denominator

To express a fraction with a complex number in the denominator in standard form, multiply the numerator and denominator by the conjugate of the denominator.

<u>ex.</u> Rationalize and write in standard form

$$\left(\frac{1}{3+2i}\right)\left(\frac{3-2i}{3-2i}\right) = \frac{3-2i}{3^2+2^2} = \frac{3-2i}{13}$$

Either FOIL or thm a^2+b^2

$$= \boxed{\frac{3}{13} - \frac{2i}{13}} = \boxed{\frac{3}{13} + \frac{-2i}{13}}$$

$$= \frac{3}{13} + \frac{-2}{13}i$$

<u>ex.</u> Perform the operation $\dfrac{1-3i}{4+2i}$ and write in standard form.

$$\left(\frac{1-3i}{4+2i}\right)\left(\frac{4-2i}{4-2i}\right) = \frac{4-2i-12i+6\overset{=-1}{i^2}}{4^2+2^2} = \frac{\overset{-1}{\cancel{-2}}-\overset{-7}{\cancel{14}}i}{\cancel{20}_{10}}$$

$$= \frac{-1-7i}{10} = \boxed{\frac{-1}{10} - \frac{7}{10}i}$$

$$\frac{\frac{1}{2}}{\frac{1}{2}}\left(\frac{-2-14i}{20}\right) = \frac{-1-7i}{10}$$

Square Roots of Negative Numbers

Def. If a is a positive real number then the **principal square root** of $-a$ is given by

Note: $\sqrt{-a} = \sqrt{-1 \cdot a}$

$$\sqrt{-a} = \sqrt{a}i = \sqrt{a} \cdot i$$

$= \sqrt{-1} \cdot \sqrt{a} = i \cdot \sqrt{a} = \sqrt{a}\, i = \sqrt{a} \cdot i$

ex. Evaluate:

1) $\sqrt{-64} = \sqrt{-1 \cdot 64} = \sqrt{-1} \cdot \sqrt{64} = i \cdot 8 = \boxed{8i}$

short-cut

2) $\sqrt{-200} = \sqrt{200}\, i = \sqrt{100 \cdot 2}\, i = \boxed{10\sqrt{2}\, i}$

NOTE: Convert the negative radicand (part under the $\sqrt{}$) of a square root to a complex number before multiplying.

($\sqrt{a}\sqrt{b} = \sqrt{ab}$ is NOT valid if both a and b are negative numbers)

ex. 1) $\sqrt{-3}\sqrt{-3} = \sqrt{3}\, i \cdot \sqrt{3}\, i = (\sqrt{3})^2 i^2 = 3(-1)$
$= \boxed{-3}$

2) $\sqrt{(-3)(-3)} = \sqrt{9} = \boxed{3}$

ex. Perform the operations and simplify:

1) $\sqrt{-8}+\sqrt{-18} = \sqrt{8}\,i + \sqrt{18}\,i =$

$2\sqrt{2}\,i + 3\sqrt{2}\,i = \boxed{5\sqrt{2}\,i}$

$\sqrt{2}\,i(2+3) = \sqrt{2}\,i \cdot 5 = 5\sqrt{2}\,i$

2) $\sqrt{-5}\sqrt{-15} = \sqrt{5}\,i \cdot \sqrt{15}\,i = \sqrt{75}\,i^2$

$= 5\sqrt{3}\,(-1)$

$= \boxed{-5\sqrt{3}}$

3) $(1+\sqrt{-2})^2 = (1+\sqrt{2}\,i)^2 =$

$(1+\sqrt{2}\,i)(1+\sqrt{2}\,i) = 1+\sqrt{2}\,i+\sqrt{2}\,i+2\underbrace{i^2}_{=-1}$

$= 1+2\sqrt{2}\,i-2$

$= \boxed{-1+2\sqrt{2}\,i}$

Complex Solutions of Quadratic Equations

In the complex number system, the solutions of the quadratic equation $ax^2+bx+c=0$, where a, b and c are real numbers and $a \neq 0$ are given by the formula

$$\begin{array}{r} 40 \\ -16 \\ \hline 24 \end{array}$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

<u>ex.</u> Solve the equation $x^2+4x+10=0$ in the complex number system. $a=1 \quad b=4 \quad c=10$

zeros

$$x = \frac{-4 \pm \sqrt{16-4(1)(10)}}{2} = \frac{-4 \pm \sqrt{-24}}{2} =$$

$$\frac{-4 \pm \sqrt{24}\, i}{2} = \frac{\cancel{-4}^{-2} \pm \cancel{2}^{1}\sqrt{6}\, i}{\cancel{2}} = \frac{-2 \pm \sqrt{6}\, i}{1}$$

$$= \boxed{-2 \pm \sqrt{6}\, i}$$

NOTE: The complex solutions are always conjugates of each other!

or $\boxed{-2+\sqrt{6}\, i}$ or $\boxed{-2-\sqrt{6}\, i}$

To summarize:

For the equation $ax^2 + bx + c = 0$ with real coefficients, find the discriminant $D = b^2 - 4ac$.

1. If $D > 0$, the equation has two unequal real number solutions.

2. If $D = 0$, the equation has a repeated real solution.

3. If $D < 0$, the equation has two complex conjugate solutions that are not real numbers.

ex. Determine the character of the solution(s) of the following equation:

$12x - 2x^2 - 18 = 0$

1) $2x(6 - x) = 18$

$-\frac{1}{2}[-2x^2 + 12x - 18 = 0]$

$x^2 - 6x + 9 = 0$

$b^2 - 4ac \rightarrow 36 - 4(1)(9) = 0$ one sol!

2) $x^{-2} - 3x^{-1} + 3 = 0$

$x^2\left(\frac{1}{x^2} - \frac{3}{x} + 3 = 0\right)$

$1 - 3x + 3x^2 = 0$

$3x^2 - 3x + 1 = 0$

$b^2 - 4ac \rightarrow$

$9 - 4(3)(1) = -3$